

DATA 615 - Regression: Homework - 5

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1 Use Handout2_SLR_b.pdf, p.8 as a summary, and read textbook Ch.4.4 – Ch.4.7 about some “other topics in regression analysis.”

Regression through origin

- $Y_i = B_1 X_i + e_i$
- Estimation equations need to be changed.
- Residuals may not sum to 0.
- ANOVA table needs adjustments (SSTO may be less than SSE).

Measurement Error

- For Y, this is usually not a problem. Just adds to the random error.
- For X, we can get biased estimators of our regression parameters.
- See text p.165 for more details.

Inverse Prediction

- Sometimes it is called calibration.
- Given Y_h , predict the corresponding value of X, by solving the fitted linear equation.
- Approximate CI can be obtained.
- See text p.169 for more details.

Choice of X levels

- Many questions: How many levels? How far between 2 levels? Range of X? How many observations at each level?
- Answer: It depends!
- See text p.171 for more details.

2 Problem 3.17 (skip b). Sales growth. (p. 150, 25pts)

Sales growth. A marketing researcher studied annual sales of a product that had been introduced 10 years ago. The data are as follows, where X is the year (coded) and Y is sales in thousands of units:

```
sg <- read.table("./data/CH03PR17.txt", header = F)
colnames(sg) <- c("sales", "year")
glimpse(sg)
```

```
## Rows: 10
## Columns: 2
## $ sales <dbl> 98, 135, 162, 178, 221, 232, 283, 300, 374, 395
## $ year <dbl> 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
```

```
summary(sg)
```

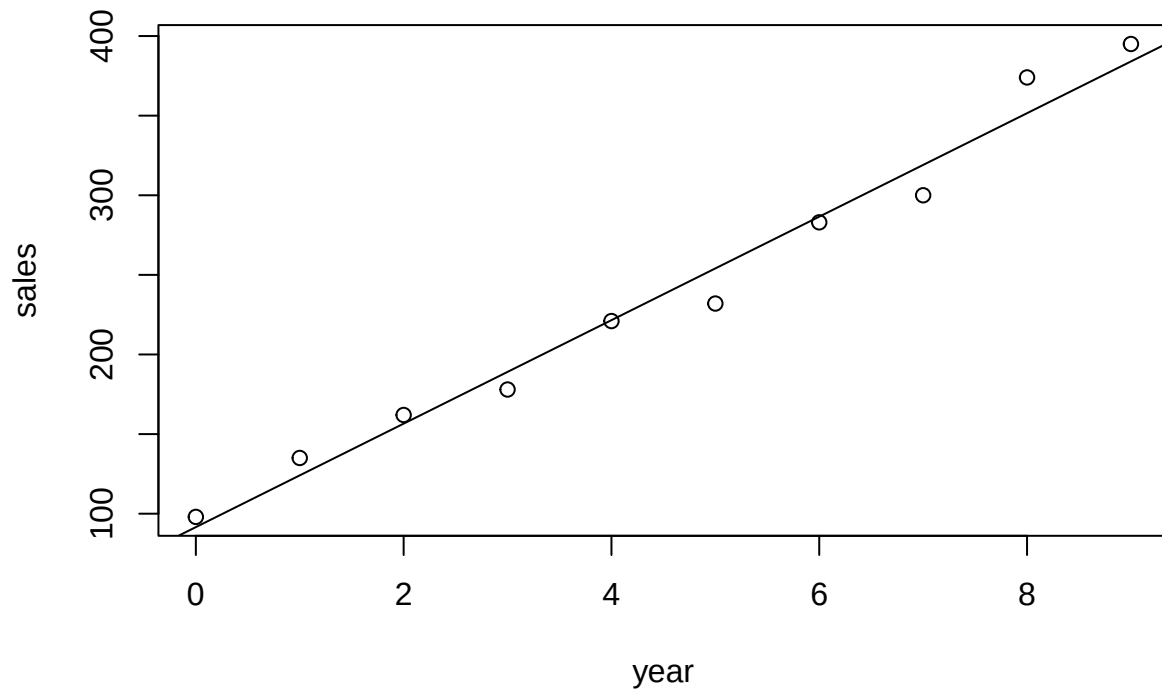
```
##      sales      year
## Min.   : 98.0   Min.   :0.00
## 1st Qu.:166.0   1st Qu.:2.25
## Median :226.5   Median :4.50
## Mean   :237.8   Mean   :4.50
## 3rd Qu.:295.8   3rd Qu.:6.75
## Max.   :395.0   Max.   :9.00
```

a. Prepare a scatter plot of the data. Does a linear relation appear adequate here?

```
sg_SLR = lm(data = sg, sales ~ year)
summary(sg_SLR)
```

```
##
## Call:
## lm(formula = sales ~ year, data = sg)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.049  -9.177   2.446   9.814  22.461
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   91.564      8.814   10.39 6.38e-06 ***
## year          32.497      1.651   19.68 4.62e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15 on 8 degrees of freedom
## Multiple R-squared:  0.9798, Adjusted R-squared:  0.9772
## F-statistic: 387.4 on 1 and 8 DF, p-value: 4.62e-08
```

```
plot(sales ~ year, data = sg)
abline(sg_SLR$coefficients)
```



Findings: Yes, the scatter plot suggests that there is a positive linear relationship between sales and year since sales increase as year increases.

b. skipped

c. Use the transformation $Y' = \sqrt{Y}$ and obtain the estimated linear regression function for the transformed data.

```
sg$salesroot = sqrt(sg$sales)
glimpse(sg)
```

```
## Rows: 10
## Columns: 3
## $ sales      <dbl> 98, 135, 162, 178, 221, 232, 283, 300, 374, 395
## $ year       <dbl> 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
## $ salesroot  <dbl> 9.899495, 11.618950, 12.727922, 13.341664, 14.866069, 15.231~
```

```
sg_SLR_2 = lm(data = sg, salesroot ~ year)
summary(sg_SLR_2)
```

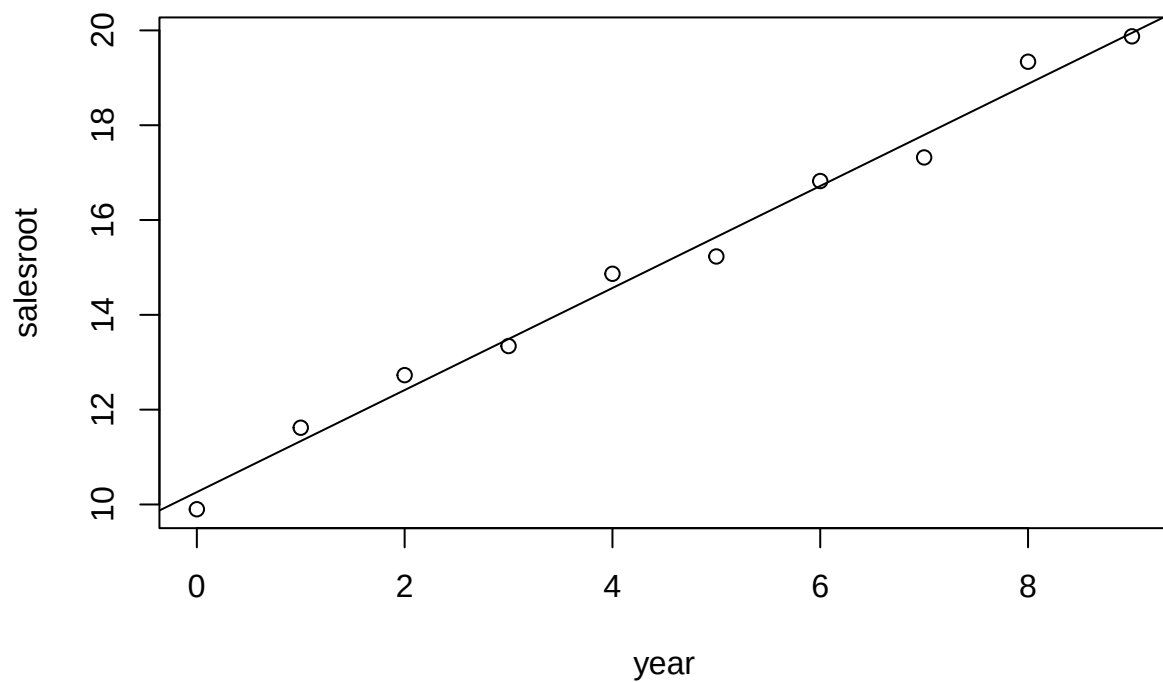
```
##
## Call:
## lm(formula = salesroot ~ year, data = sg)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.47447 -0.30811  0.01549  0.29541  0.46781
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 10.26093    0.21290   48.20 3.80e-11 ***
## year         1.07629    0.03988   26.99 3.83e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3622 on 8 degrees of freedom
## Multiple R-squared:  0.9891, Adjusted R-squared:  0.9878
## F-statistic: 728.4 on 1 and 8 DF,  p-value: 3.826e-09
```

The estimated linear regression function for this model is:

$$\hat{Y} = 10.26 + 1.076X$$

d. Plot the estimated regression line and the transformed data. Does the regression line appear to be a good fit to the transformed data?

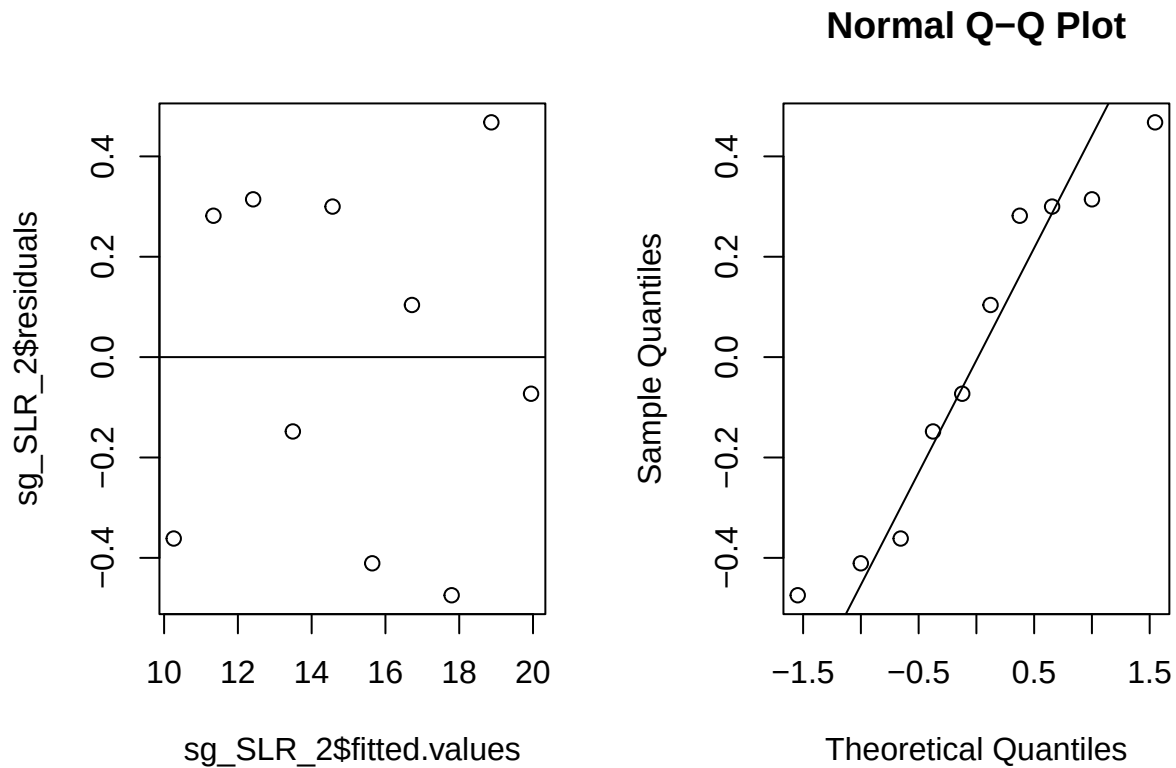
```
plot(salesroot ~ year, data = sg)
abline(sg_SLR_2$coefficients)
```



Findings: Yes, the regression line does seem to be a good fit to the **transformed** sales variable, with all data points falling very closely the the regression line.

e. Obtain the residuals and plot them against the fitted values. Also prepare a normal probability plot. What do your plots show?

```
par(mfrow = c(1, 2))
plot(sg_SLR_2$residuals ~ sg_SLR_2$fitted.values)
abline(0, 0)
qqnorm(sg_SLR_2$residuals)
qqline(sg_SLR_2$residuals)
```



- **Findings:** The residual plot does seem evenly distributed, but it might suggest that the variance isn't constant as it seems to increase along with the estimated Ys. The QQ-plot also suggests that the data doesn't seem to be normally distributed since the data points don't fall along the reference line as expected. However, this data set is very small with only 10 observations and not suitable to run the Shaphiro-Wilk test for normality since the X values are unique.

f. Express the estimated regression function in the original units.

The final estimated model is: $\sqrt{\hat{Y}} = 10.26 + 1.076X$.

Expressing the estimated regression model in its original scale:

$$\hat{Y} = (10.26 + 1.076X)^2$$

3 Problem 4.7. Copier maintenance (p. 172, 20 pts. a-10, b-5, c-5)

Observations:

- Review your work in HW 2, Problem 2.14 (a, b) for the confidence interval of the mean response (aka. the expected value of the response) and the prediction interval of an individual new response.
- Please read the examples about the critical values for simultaneous inference in Lab 3, p.6, 7, or in handout FamilyCI.pdf
- In part (a), use both Working-Hotelling and Bonferroni to construct the confidence interval. Clearly state the critical values used in each procedure. Which procedure produces narrower confidence interval?
- In part (b), answer the question without computing the actual PI. (Hint: find the critical values.)
- In part (c), use both Scheffe and Bonferroni to construct the prediction intervals.
- In both (a) and (c), you can easily confirm your results for Bonferroni procedure using software.

Variables description for the considered data file:

time: Y is the total number of minutes spent by the service person

copiers: X is the number of copiers serviced

copiers_age: X2 is the mean operational age of copiers serviced on the call (in months).

workers_experience: X3 is number of years of experience of the service person making the call.

```
cm <- read.table("./data/CH03PR04.txt", header = F)
colnames(cm) <- c("time", "copiers", "copiers_age", "workers_experience")
glimpse(cm)
```

```
## Rows: 45
## Columns: 4
## $ time      <int> 20, 60, 46, 41, 12, 137, 68, 89, 4, 32, 144, 156, 9~
## $ copiers    <int> 2, 4, 3, 2, 1, 10, 5, 5, 1, 2, 9, 10, 6, 3, 4, 8, 7~
## $ copiers_age <int> 20, 19, 27, 32, 24, 26, 24, 28, 22, 18, 25, 25, 32, ~
## $ workers_experience <int> 4, 5, 4, 1, 4, 4, 3, 6, 6, 6, 4, 4, 6, 3, 6, 3, 4, ~
```

```
summary(cm)
```

##	time	copiers	copiers_age	workers_experience
##	Min. : 3.00	Min. : 1.000	Min. : 9.00	Min. :1.000
##	1st Qu.: 36.00	1st Qu.: 2.000	1st Qu.:21.00	1st Qu.:3.000
##	Median : 74.00	Median : 5.000	Median :25.00	Median :4.000
##	Mean : 76.27	Mean : 5.111	Mean :24.64	Mean :3.978
##	3rd Qu.:111.00	3rd Qu.: 7.000	3rd Qu.:28.00	3rd Qu.:5.000
##	Max. :156.00	Max. :10.000	Max. :36.00	Max. :7.000

```
cm_SLR <- lm(data = cm, time ~ copiers)
summary(cm_SLR)
```

```
##
## Call:
## lm(formula = time ~ copiers, data = cm)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.7723  -3.7371   0.3334   6.3334  15.4039
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.5802     2.8039  -0.207   0.837
## copiers       15.0352     0.4831  31.123 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.914 on 43 degrees of freedom
## Multiple R-squared:  0.9575, Adjusted R-squared:  0.9565
## F-statistic: 968.7 on 1 and 43 DF,  p-value: < 2.2e-16
```

The estimated regression function is:

$$\widehat{time} = -0.5802 + 15.0352(copiers)$$

a. Estimate the expected number of minutes spent when there are 3, 5, and 7 copiers to be serviced, respectively. Use interval estimates with a 90 percent family confidence coefficient based on the Working-Hotelling procedure.

- Objective: estimate the expected number of minutes spent on servicing when $X = (3, 5, 7)$.
- The family confidence coefficient is 90%.
- Family $\alpha = 0.01$.
- $g = 3$.
- $n = 45$.
- For the Work-Hotelling's procedure:

$$- \text{CI: } \hat{y} + / - W * se(\hat{y}_{mean})$$

```
W = sqrt(2 * qf(0.9, 2, 43))
W
```

```
## [1] 2.204725
```

- $W = \sqrt{2F_{(1-\alpha;2;n-2)}} = \sqrt{2F_{(0.9;2;43)}} = 2.204725$

```
predict(cm_SLR,
        newdata = data.frame(copiers=c(3, 5, 7)),
        se = T,
        level = 1 - (0.1/3),
        interval = "confidence") -> cm_CI_Yhat
```

```
cm_CI_Yhat
```

```
## $fit
##      fit      lwr      upr
## 1  44.52559  40.84285  48.20832
## 2  74.59608  71.67227  77.51989
## 3 104.66658 101.12260 108.21056
##
## $se.fit
##      1      2      3
## 1.675012 1.329831 1.611900
##
## $df
## [1] 43
##
## $residual.scale
## [1] 8.913508
```

```
Y_hat = cm_CI_Yhat$fit[, 1]
```

```
#Y_hat
```

```
se_Y_hat = cm_CI_Yhat$se.fit
```

```
#se_Y_hat
```

```
lower_CI = Y_hat - W * se_Y_hat
```

```
#lower_CI
```

```
upper_CI = Y_hat + W * se_Y_hat
```

```
#upper_CI
```

```
data.frame(X = c(3, 5, 7), Y_hat, se_Y_hat, lower_CI, upper_CI)
```

```
##   X    Y_hat se_Y_hat lower_CI upper_CI
## 1 3  44.52559 1.675012  40.83265  48.21853
## 2 5  74.59608 1.329831  71.66417  77.52800
## 3 7 104.66658 1.611900 101.11278 108.22038
```

Findings:

At 90% confidence level using the Work-Hotelling method, for $X = 3$, the CI for mean service time is (40.83265, 48.21853), for $X = 5$, it is (71.66417, 77.52800) and, for $X = 7$, it is (101.11278, 108.22038).

b. Two service calls for preventive maintenance are scheduled in which the numbers of copiers to be serviced are 4 and 7, respectively. A family of prediction intervals for the times to be spent on these calls is desired with a 90 percent family confidence coefficient. Which procedure, Scheffe or Bonferroni, will provide tighter prediction limits here?

- Objective: estimate a family of prediction intervals for the times to be spent on these calls.

- The family confidence coefficient is 90%.
- Family alpha = 0.01.
- $g = 2$.
- $n = 45$.
- For the Bonferroni's procedure: $B = t_{(1-\alpha/2g, df = df_E = (n-2))} = t_{(1-(0.1/4), 43)} = 2.016692$

```
B <- qt(1 - 0.1/(2*2), 43)
B
```

```
## [1] 2.016692
```

- For the Scheffe procedure: $S = \sqrt{gF_{(1-\alpha; g; n-2)}} = \sqrt{2F_{(0.9; 2; 43)}} = 2.204725$

```
S = sqrt(2 * qf(0.9, 2, 43))
S
```

```
## [1] 2.204725
```

Answer: Since the t-critic value calculated from the Bonferroni's procedure is smaller than the one from the Scheffe's procedure, the Bonferroni method will allow for a tighter prediction intervals.

c. Obtain the family of prediction intervals required in part (b), using the more efficient procedure.

The Bonferroni's procedure can be calculated directly using the predict() function below:

```
predict(cm_SLR,
        newdata = data.frame(copiers=c(4, 7)),
        se = T,
        level = 1 - (0.1/2),
        interval = "prediction")
```

```
## $fit
##      fit      lwr      upr
## 1  59.56084 41.35419 77.76748
## 2 104.66658 86.39922 122.93394
##
## $se.fit
##      1      2
## 1.433068 1.611900
##
## $df
## [1] 43
##
## $residual.scale
## [1] 8.913508
```

Findings:

At 90% confidence level using the Bonferroni's method, for $X = 4$, the PI for times to be spent on calls will be (41.35419, 77.76748) and, for $X = 7$, it will be (86.39922, 122.93394).

4 Problem 6.9. Grocery Retailer (p. 249, 15 pts.)

Observations:

- In part (a), you can use histogram instead of stem-and-leaf plot.
- In part (b), since the observations (aka. cases, units, etc.) are time-ordered (“given in consecutive weeks”), the “time plot” is just a “sequence plot” or “index plot.”
 - In R, use the following function (“l” is the lower case of letter L) for an index (or sequence) plot of “variable” in data frame “gr.data”: (you may name your data frame and variables differently.)
`plot(gr.data$variable, type="l")`
 - If there is a time variable, you can use a scatter plot or line plot, with the time variable in the horizontal axis.
 - If there is a time variable, you can use a scatter plot or line plot, with the time variable in the horizontal axis.
- In part (c), the word “matrix” means to organize the pair-wise plots and the correlation coefficients in an array.
 - R users: see example in RLab3_Multiple1.pdf, p.3.

A large, national grocery retailer tracks productivity and costs of its facilities closely. Data below were obtained from a single distribution center for a one-year period. Each data point for each variable represents one week of activity.

The variables included are:

- The total labor hours (Y).
- The number of cases shipped (X1),
- The indirect costs of the total labor hours as a percentage (X2),
- A qualitative predictor called holiday that is coded 1 if the week has a holiday and 0 otherwise (X3),

```
gr <- read.table("./data/CH06PR09.txt", header = F)
colnames(gr) <- c("Y_labor_hours", "X1_cases_shipped", "X2_cost_labor", "X3_holiday_dummy")
glimpse(gr)
```

```
## Rows: 52
## Columns: 4
## $ Y_labor_hours    <int> 4264, 4496, 4317, 4292, 4945, 4325, 4110, 4111, 4161, ~
## $ X1_cases_shipped <int> 305657, 328476, 317164, 366745, 265518, 301995, 26933~
## $ X2_cost_labor    <dbl> 7.17, 6.20, 4.61, 7.02, 8.61, 6.88, 7.23, 6.27, 6.49, ~
## $ X3_holiday_dummy <int> 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, ~
```

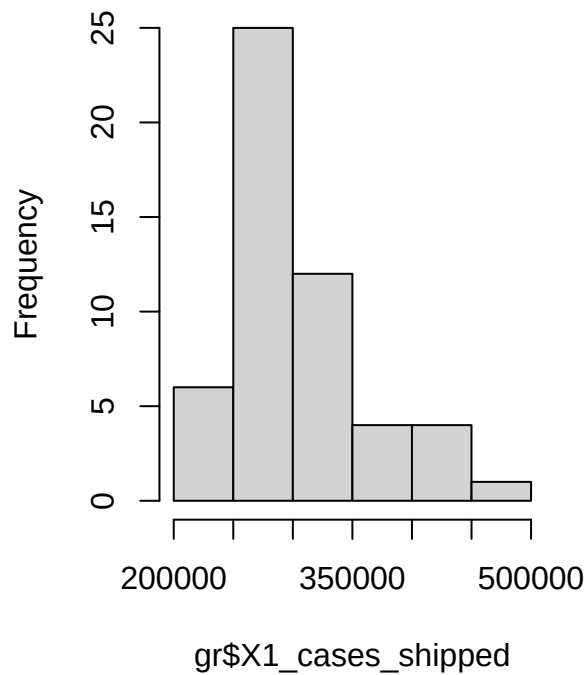
```
summary(gr)
```

```
## Y_labor_hours X1_cases_shipped X2_cost_labor X3_holiday_dummy
## Min. :3998 Min. :211944 Min. :4.610 Min. :0.0000
## 1st Qu.:4193 1st Qu.:268759 1st Qu.:6.805 1st Qu.:0.0000
## Median :4316 Median :291271 Median :7.325 Median :0.0000
## Mean :4363 Mean :302693 Mean :7.371 Mean :0.1154
## 3rd Qu.:4472 3rd Qu.:321906 3rd Qu.:7.938 3rd Qu.:0.0000
## Max. :5045 Max. :472476 Max. :9.650 Max. :1.0000
```

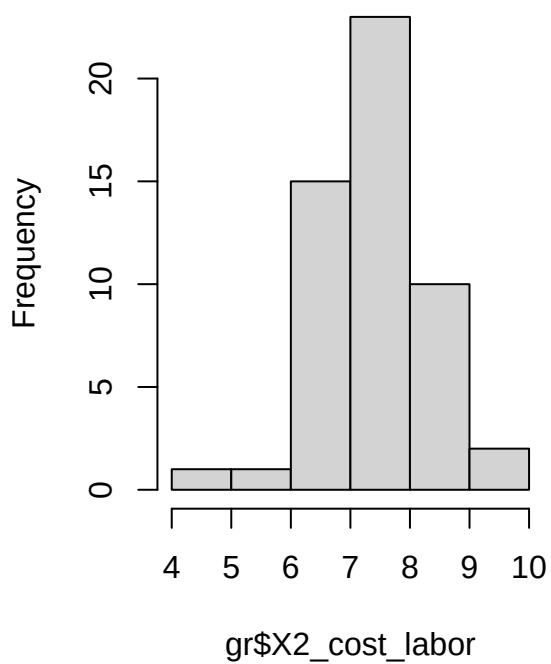
a. Prepare separate stem-and-leaf plots for the number of cases shipped X1 and the indirect cost of the total hours X2. Are there any outlying cases present? Are there any gaps in the data?

```
par(mfrow = c(1, 2))
hist(gr$X1_cases_shipped)
hist(gr$X2_cost_labor)
```

Histogram of gr\$X1_cases_shipp



Histogram of gr\$X2_cost_labor

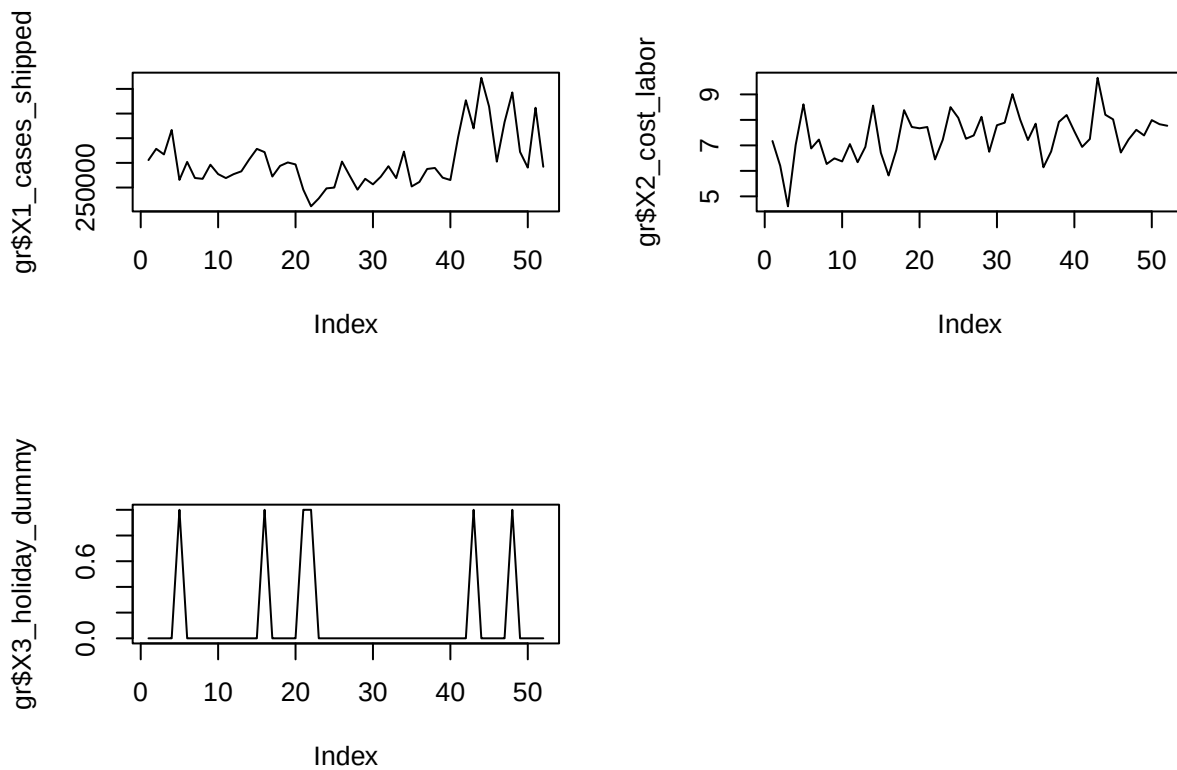


- The variable 'number of cases shipped' appears to be slightly skewed to the right and there's no visible outliers. The Variable 'cost of labor hours' appears to be skewed to the left and also with no visible outliers. There are no gaps in the data.

b. The cases are given in consecutive weeks. Prepare a time plot for each predictor variable. What do the plots show?

```
par(mfrow = c(2, 2))

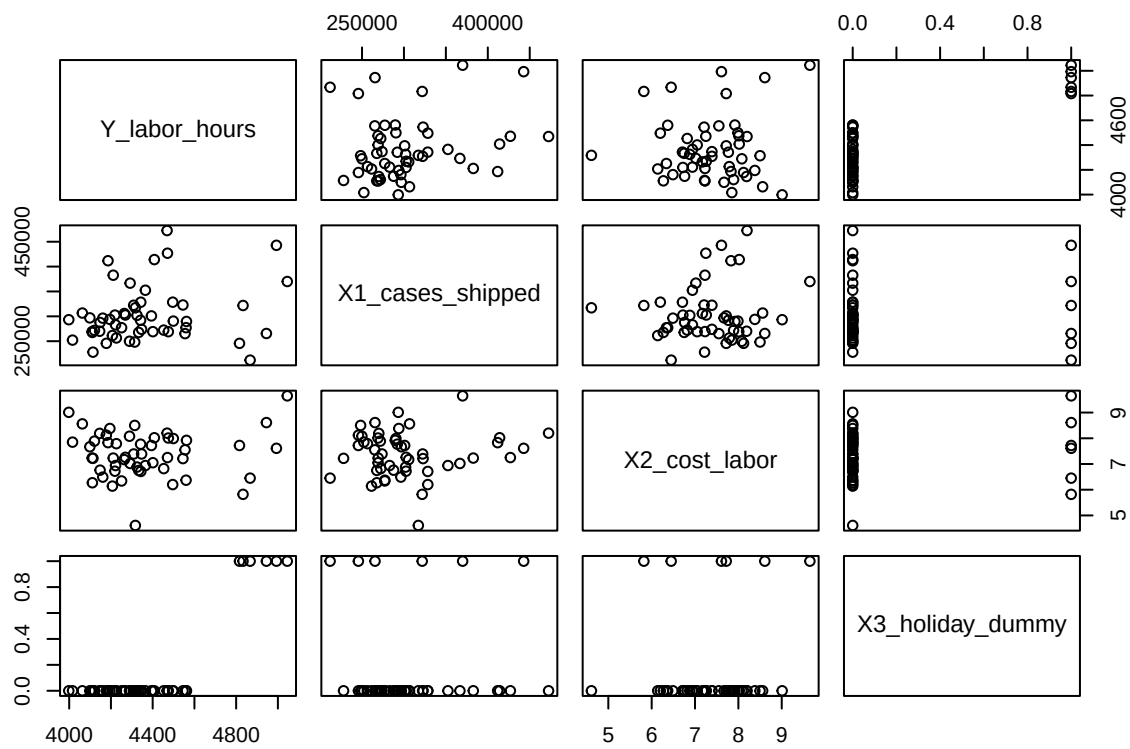
plot(gr$X1_cases_shipped, type = "l")
plot(gr$X2_cost_labor, type = "l")
plot(gr$X3_holiday_dummy, type = "l")
```



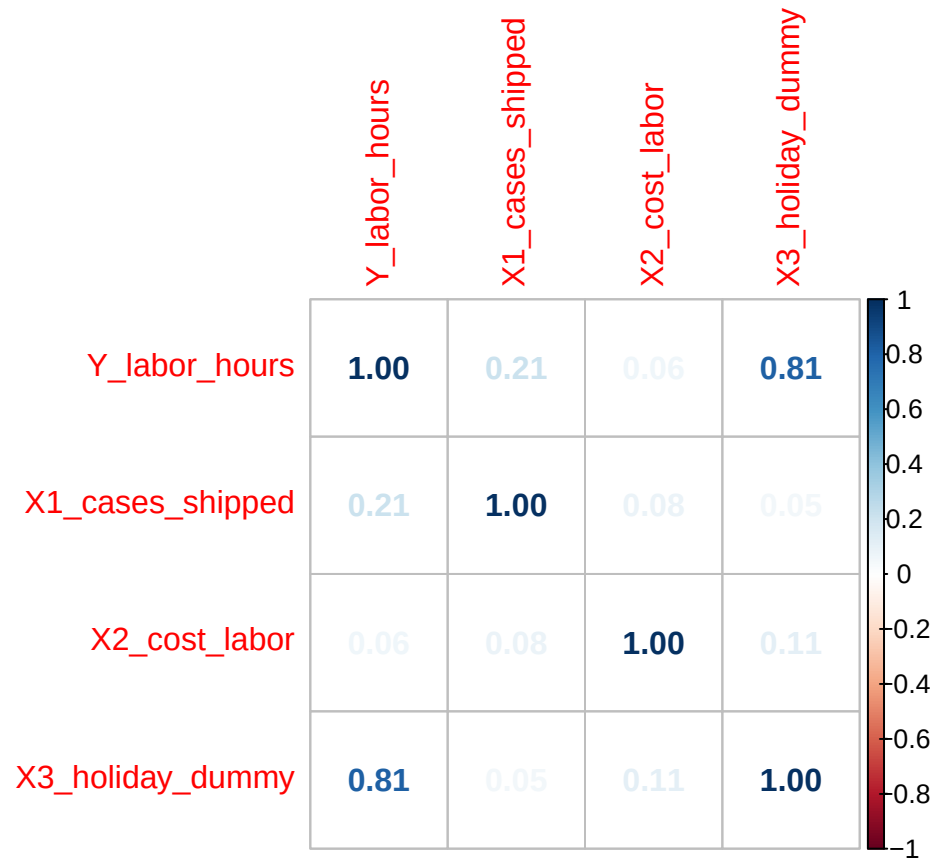
Findings: There is clear patterns in either of the graphs above. However, the ‘number of cases shipped’ variable does seem to increase with time. The X3 variable is erratic because it a dummy, 0 or 1, variable.

c. Obtain the scatter plot matrix and the correlation matrix. What information do these diagnostic aids provide here?

```
pairs(gr)
```



```
cor_matrix <- round(cor(gr, use = "complete.obs"), 2)
corrplot(cor_matrix, method="number")
```

Findings:

- The scatter plot matrix shows that the only visible association is between X3 (holiday indicator) and Y (labor hours), which can be confirmed with the correlation matrix above. All the other variables show a weak correlation between each other.

5 Problem 6.10. (a, b, c, d) (p. 249, 25 pts: a, b, c-10, d-5)

Observations:

- In part (a), the question referred to model (6.5) on p.217. In this case, the model is: $Y = B_0 + B_1X_1 + B_2X_2 + B_3X_3 + e$. The interpretation of the B's and their estimates (b_0, b_1, b_2, b_3) are the same as the interpretation in simple linear regression, except a statement to clarify that the slope is estimated "....., after controlling (or adjusted for) the association (or effect) of other predictors." You may also use the language "....., while holding the other predictors constant."
- In part (b), you can use either histogram or box plot.
- In part (c), you will create 5 residual plots, and a Normal Q-Q plot for the residuals. Be sure to summarize your findings from the plots. To plot the residuals (from part a) against X_1X_2 , you can create a new variable, say $X_4 = X_1X_2$, and plot the residuals (from part a) against X_4 . This plot will suggest whether X_4 should be added to the regression model.

a. Fit regression model (6.5) to the data for three predictor variables. State the estimated regression function. How are b_1, b_2 and b_3 interpreted here?

```
gr_MLR = lm(Y_labor_hours ~ X1_cases_shipped + X2_cost_labor + X3_holiday_dummy, data = gr)
summary(gr_MLR)
```

```
##
## Call:
## lm(formula = Y_labor_hours ~ X1_cases_shipped + X2_cost_labor +
##     X3_holiday_dummy, data = gr)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -264.05 -110.73  -22.52   79.29  295.75
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.150e+03  1.956e+02  21.220 < 2e-16 ***
## X1_cases_shipped  7.871e-04  3.646e-04   2.159  0.0359 *
## X2_cost_labor   -1.317e+01  2.309e+01  -0.570  0.5712
## X3_holiday_dummy  6.236e+02  6.264e+01   9.954 2.94e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 143.3 on 48 degrees of freedom
## Multiple R-squared:  0.6883, Adjusted R-squared:  0.6689
## F-statistic: 35.34 on 3 and 48 DF,  p-value: 3.316e-12
```

- The estimated regression function is:

$$\hat{Y} = 4150 + 0.000787X_1 - 13.17X_2 + 623.6X_3$$

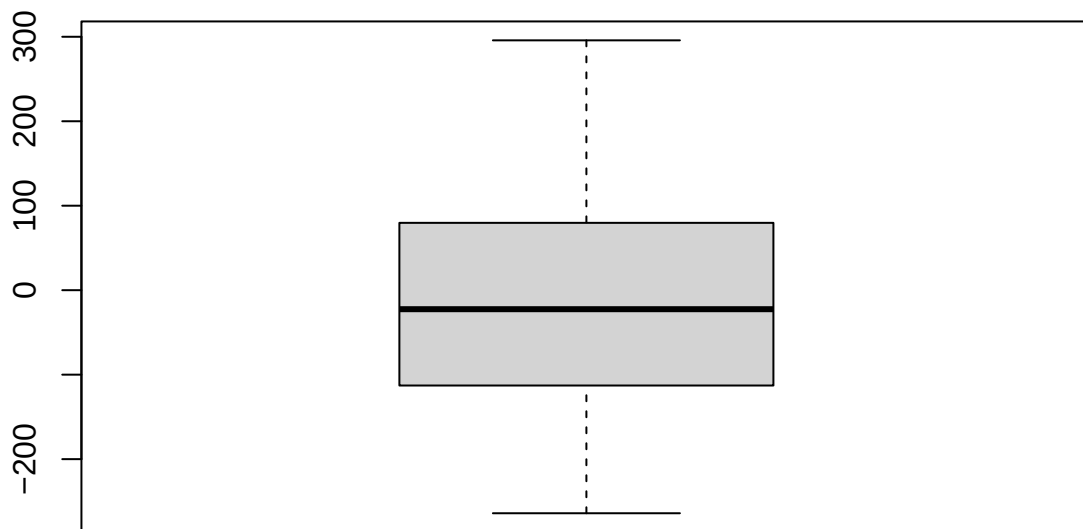
- the b1, b2 and b3 are the estimated slope coefficients for the variables X1, X2 and X3, respectively.
- Their interpretations are as following:
 - For variable X1_cases_shipped: For each one unit increase in cases shipped, the mean total labor hours (Y) is expected to increase by 0.000787, holding all the other predictors constant.
 - For variable X2_cost_labor: For each one unit increase in labor cost, the mean total labor hours (Y) is expected to decrease by 13.17, holding all the other predictors constant.
 - For variable X3_holiday_dummy: When there is a holiday on a given week, the mean total labor hours (Y) is expected to increase by 623.6, holding all the other predictors constant.

b. Obtain the residuals and prepare a box plot of the residuals. What information does this plot provide?

```
gr_MLR$residuals
```

```
##          1          2          3          4          5          6
## -32.063483 169.205091 -21.825426 -54.119552  75.933724  28.000660
##          7          8          9         10         11         12
## -156.684401 -166.983381 -136.691019  275.783546  132.059842 -33.540598
##          13         14         15         16         17         18
## -59.173782 -215.535629  22.975644 -117.076677  178.568096 -75.863154
##          19         20         21         22         23         24
##  108.947943 -183.565972 -49.165210  11.662167 -120.279732  80.569854
##          25         26         27         28         29         30
##   48.807558 -23.519661  78.869281 -58.398630  61.303251 -23.214763
##          31         32         33         34         35         36
## -138.978271 -264.053024  218.752742  235.960794 -229.055311 -67.763118
##          37         38         39         40         41         42
## -139.284659  288.397234 -108.609359  295.751820  29.065887  80.555515
##          43         44         45         46         47         48
##  107.399309  55.197555  37.772702 -80.508888 -144.901536 -28.753312
##          49         50         51         52
##   2.731302  225.697849 -184.877629  64.516810
```

```
boxplot(gr_MLR$residuals)
```



Findings: Based on the box plot, the residuals seem to be evenly distributed around the mean zero and with no visible outliers.

c. Plot the residuals against $\hat{Y}$, X_1 , X_2 , X_3 , and X_1X_2 on separate graphs. Also prepare a normal probability plot. Interpret the plots and summarize your findings.

```
par(mfrow = c(2, 3))

plot(gr_MLR$fitted.values, gr_MLR$residuals)
abline(0, 0)

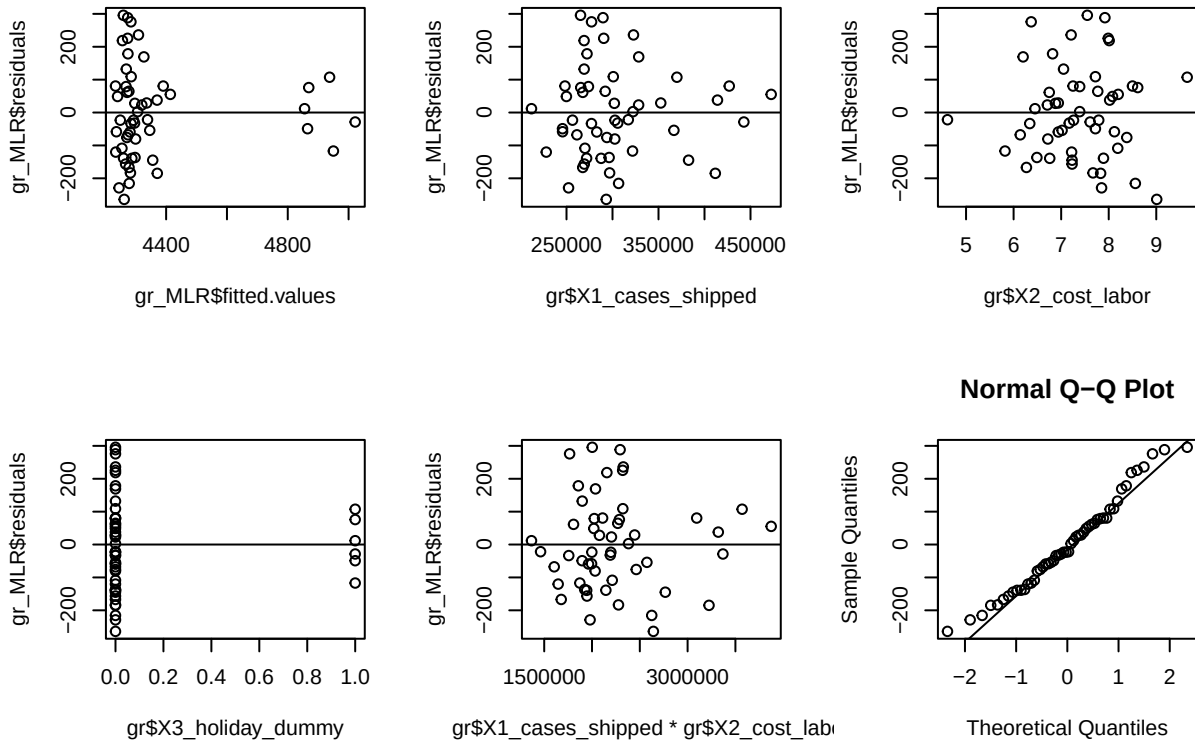
plot(gr$X1_cases_shipped, gr_MLR$residuals)
abline(0, 0)

plot(gr$X2_cost_labor, gr_MLR$residuals)
abline(0, 0)

plot(gr$X3_holiday_dummy, gr_MLR$residuals)
abline(0, 0)

plot(gr$X1_cases_shipped*gr$X2_cost_labor, gr_MLR$residuals)
abline(0, 0)

qqnorm(gr_MLR$residuals)
qqline(gr_MLR$residuals)
```

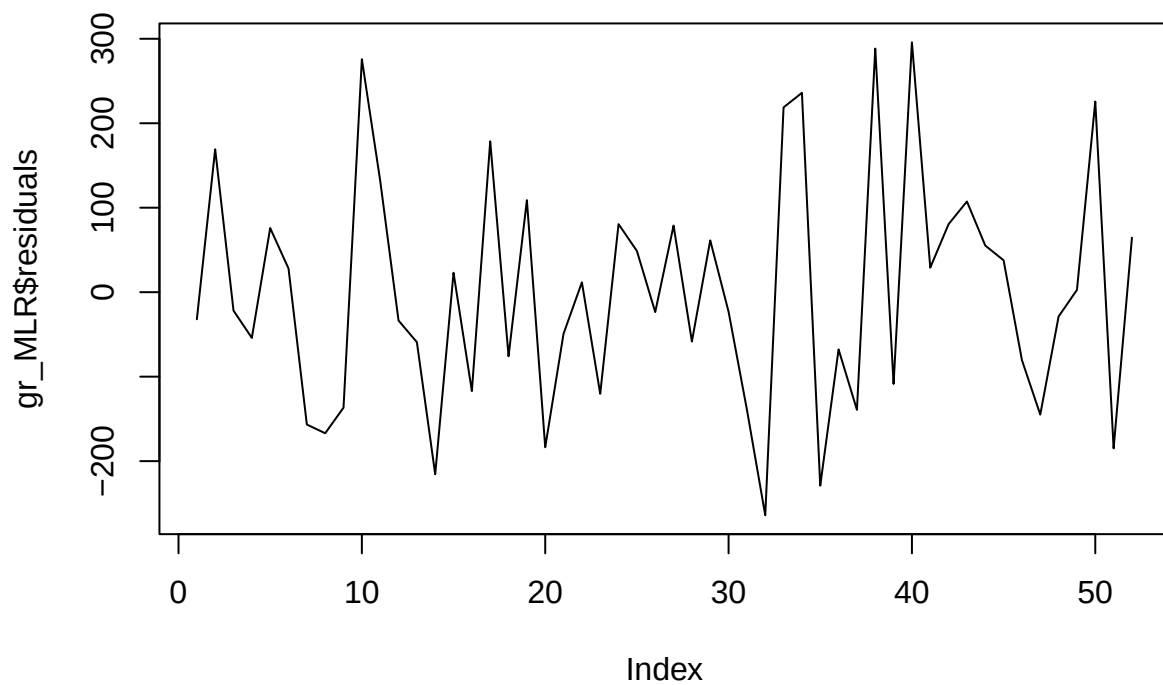


Findings:

- The plot of residuals vs. the fitted values shows presence of varying variance and possible outliers from some few observations separated from the rest.
- The plot of residual vs. X1 also shows varying variance as number of cases shipped increases.
- The plot of residual vs. X2 shows that the data points are more evenly distributed and with no obvious patterns.
- The plot of residual vs. X2 also shows varying variance for the non-holiday indicator.
- The plot of residual vs. X1*X2 is very similar to the plot of residual vs. X1, presenting varying variance and clear patterns.
- The normality plot doesn't seem to indicate any problems with the residuals and we can assume the residuals are normally distributed.

d. Prepare a time plot of the residuals. Is there any indication that the error terms are correlated? Discuss.

```
plot(gr_MLR$residuals, type = "l")
```



Findings: The time plot of the residuals doesn't show any specific pattern and no evidence of out correlation of the residuals over time.

‘The End‘